

Using Distances and Pair Counting Correlations to Estimate Hierarchical Density-Based Clustering Parameters

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Abstract. Unsupervised clustering is essential for exploring large unlabeled datasets, yet hierarchical density-based methods often suffer from over-fragmentation in high-dimensional semantic spaces. This paper introduces **Core-HDBSCAN**, a deterministic approach for estimating spatial thresholds (ϵ^*) using fractal correlation analysis. By measuring distance correlations between data subsets, the method identifies macro-structural boundaries separating semantic manifolds and derives a geometrically grounded threshold for dendrogram pruning. Experimental results demonstrate improved clustering calibration, reducing parameter tuning while achieving F1-score gains of up to 37% over the baseline and better preserving the underlying domain structure.

Keywords: Hierarchical Density-Based Clustering · Fractal Correlation Analysis · Spatial Threshold Estimation.

1 Introduction

As fully labeled datasets become increasingly difficult to obtain and process, unsupervised clustering has emerged as a key tool for exploring large unlabeled collections. HDBSCAN is particularly appealing because it can generate clusters using only a minimum cluster size parameter. However, meaningful results often still require iterative parameter tuning. This study addresses that limitation by deterministically estimating the hierarchical spatial threshold (ϵ^*), promoting well-structured partitions that capture broad semantic categories without excessive fragmentation into specialized subtopics. While previous methods, such as the hybrid approach of Malzer and Baum [9], attempts to identify an optimal cut-off value, their effectiveness tends to diminish in high-dimensional spaces.

To address this limitation, we introduce **Core-HDBSCAN**, a methodology that estimates the clustering threshold directly from high-dimensional semantic spaces using fractal correlation analysis. By exploiting the scaling behavior of pairwise spatial correlations, the fractal dimension reveals boundaries between distinct manifolds, providing a geometrically grounded threshold that mitigates

the dimensionality curse [7] affecting conventional density-based heuristics. The proposed technique can be applied to a wide range of datasets. For validation, we used the PIESP dataset, a collection of embeddings derived from investment-related news articles and organized into macroeconomic sectors and subsectors (Section 4). Extensively refined and used by a government institution to support public policy for several years, the dataset provides a reliable ground truth for evaluating the proposed approach.

The remainder of this paper is organized as follows. Section 2 reviews the related work and theoretical foundations. Section 3 presents the proposed fractal-correlation-based methodology. Section 4 describes the experimental setup, dataset, evaluation metrics, and results analysis. Finally, Section 5 concludes the paper and discusses future research directions.

2 Background and Related Work

Clustering is an unsupervised learning technique for discovering latent patterns and grouping similar instances in unlabeled data. *K*-Means is widely used for its simplicity and efficiency but requires a predefined number of clusters, assumes spherical cluster shapes, and is sensitive to noise [6]. For noisier datasets with arbitrary shapes, density-based approaches like DBSCAN are preferred. It identifies clusters as dense regions, detects arbitrary geometries, and filters out noise [3] using two parameters: the neighborhood radius ϵ and the minimum number of neighbors required (`MinPts`). To overcome DBSCAN’s sensitivity to ϵ and its difficulty with varying densities, extensions have been proposed. OPTICS addresses density variations by ordering points to reveal multi-level cluster structures [1]. HDBSCAN further generalizes this by constructing a density hierarchy and extracting the most stable clusters [2] using three parameters: `min_cluster_size`, `min_samples` (for noise classification), and `cluster_selection_epsilon` (to limit over-fragmentation).

HDBSCAN incorporates the GLOSH outlier score to identify points deviating from dense regions, enabling unsupervised outlier detection. While refinements have improved its robustness and reduced parameter sensitivity [4], extracting meaningful clusters still depends on density-bounding parameters like `cluster_selection_epsilon`. To prevent HDBSCAN from over-fragmenting clusters in highly dense regions, Malzer and Baum [9] proposed a hierarchical threshold parameter ($\hat{\epsilon}$) within the FOSC framework. This threshold acts as a spatial floor that halts cluster divisions below a specified distance. However, selecting an appropriate value remains challenging in high-dimensional spaces where empirical heuristics fail, particularly in semantic embedding spaces where geometric distances often distort the underlying topological structure.

Other researchs characterize datasets by analyzing the relationship between pairwise distances and neighborhood growth within a given radius, a multiscale property formalized as the intrinsic dimensionality (D). While statistical [5] and fractal-based [8] methods have been proposed to estimate D , most provide only global or local estimates, failing to explicitly model multiple clusters.

The TriPlot technique addresses this limitation by analyzing two datasets embedded in a common space, possibly with distinct spatial distributions [10]. It examines how neighborhood counts grow with distance, capturing both intra- and inter-cluster structures across scales. To isolate inter-cluster behavior, TriPlot employs the cross-cloud function, which measures the cumulative number of cross-pairs as the radius increases. The resulting log-log cross-plot provides an estimate of the intrinsic dimensionality (D) through its average slope. Figure 1 illustrates a TriPlot for the PIESP dataset (Section 4.1). The dotted curves represent the neighborhood growth within the “Industries” and “Services” subsets, while the solid blue curve depicts their cross-group interaction. The average slope of each curve corresponds to D , with differences reflecting variations in the underlying spatial distributions.

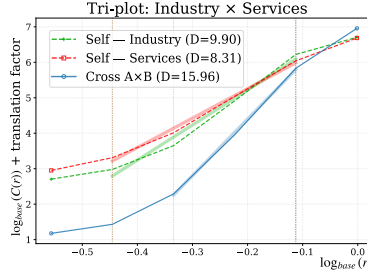


Fig. 1. Cross-plot of topological interaction between "Industry" and "Services news".

In this paper, we investigate how TriPlots-based analysis can be used to derive principled estimates for the ϵ parameter in HDBSCAN. Our objective is to replace heuristic or ad hoc parameter selection with a data-driven strategy, yielding a more reliable and stable configuration of the algorithm.

3 The Core-HDBSCAN Technique

When using HDBSCAN, practitioners typically tune `min_cluster_size` and `min_samples` parameters, while leaving `cluster_selection_epsilon` at its default value due to the difficulty of estimating it. This work introduces Core-HDBSCAN, a data-driven approach that leverages distance and pair-count correlations to estimate intrinsic dimensionality. By analyzing topological acceleration in the Cross-Cloud Plot, Core-HDBSCAN derives an informed spatial threshold (ϵ^*), enabling systematic calibration of `cluster_selection_epsilon` and reducing reliance on trial-and-error tuning in multiscale datasets.

Our heuristic models the interaction between high-dim textual embedding sets $\mathcal{A}$ and $\mathcal{B}$ across distinct semantic domains. The dissimilarity between elements $\mathbf{a} \in \mathcal{A}$ and $\mathbf{b} \in \mathcal{B}$ is defined as $r(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|_2$, with their structural relationship characterized by the cross-cloud correlation function $\text{Cross}_{\mathcal{A}, \mathcal{B}}(r)$. Core-HDBSCAN assumes a coherent macrostructure emerges at a characteristic scale, identified by the elbow (maximum acceleration) on a log-log Cross-plot.

Let $x = \log_{base}(r)$ denote the spatial scale (base 10 in our experiments). Since the slope of the log-log Cross-plot approximates the local correlation dimension, structural transitions are identified through the curvature of the cross-cloud curve, estimated via finite differences:

$$\frac{d^2}{dx^2} \log_{base}(\text{Cross}_{\mathcal{A},\mathcal{B}}(base^x)). \quad (1)$$

The characteristic interaction distance is defined as the scale of maximum curvature within the informative region of the curve, excluding the final plateau.

Definition 1 (Optimal Fractal Threshold). *Let $X_{\text{valid}} \subset X_{\text{domain}}$ denote the range preceding the final plateau of the cross-cloud curve. The optimal spatial threshold is defined as $\epsilon^* = base^{x_{\text{ref}}}$, where*

$$x_{\text{ref}} = \arg \max_{x \in X_{\text{valid}}} \left(\frac{d^2}{dx^2} \log_{base}(\text{Cross}_{\mathcal{A},\mathcal{B}}(base^x)) \right). \quad (2)$$

Using ϵ^* as a scale-selective filter, Core-HDBSCAN prevents the clustering algorithm from prematurely isolating dense sub-clusters at microscales ($r < \epsilon^*$).

3.1 Deriving the Macro-Structural Boundary using Fractal Scaling

Analyzing the cross-correlation curve, we obtain the structural properties of the data distribution in the distance space induced by cross-domain interactions. By identifying internal subclusters and global macrostructure boundaries, the following properties and rule are derived:

Property 1 (Homogeneous Correlation). A straight cross-plot curve with a single constant slope and no intermediate plateaus indicates a homogeneous structural relationship, and the analyzed dataset pair contains no macroscopic subclusters.

Property 2 (Heterogeneity and Subclusters). Conversely, varying slopes and plateaus indicate a heterogeneous relationship with subclusters. A sharp increase in steepness marks the onset of exponential pair-counting growth, while the final top plateau reveals the maximum spatial distance between the datasets.

Rule 1 (Macro-structural Extraction): To identify the entire, cohesive dataset rather than isolated subclusters, locate the start of the steepest slope. If multiple slopes exist, the last steep slope before the final saturation plateau represents the complete macroscopic correlation between the sets.

3.2 Determining the optimal clustering threshold

To illustrate the proposed method, we extract ϵ^* step-by-step using cross-plots from the PIESP dataset pairs (economic sectors). Figure 2 shows the cumulative cross-pair distribution against the expanding spatial radius r (in log scale). The evaluated bounds define the clustering limits. The lower bound is the minimum inter-dataset distance. Setting ϵ here causes severe micro-cluster fragmentation,

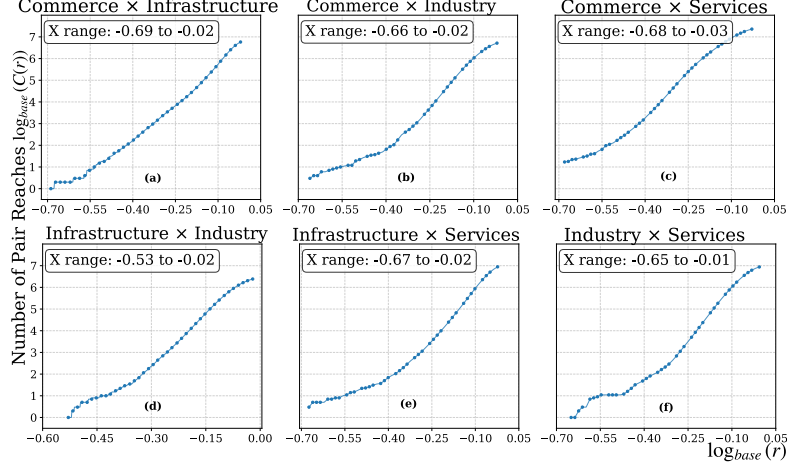


Fig. 2. Cross-correlation analysis (log-log scale) after initial and final plateau removal, spotting regions of linear growth in the cumulative inter-cluster pair distances.

treating most data as noise. The upper bound is the maximum distance before the final saturation plateau. Setting ϵ here merges distinct economic sectors into a single cluster, obscuring domain boundaries. As these curves lack a purely homogeneous distribution (as defined in Property 1), a detailed topological breakdown is required to find the optimal scale.

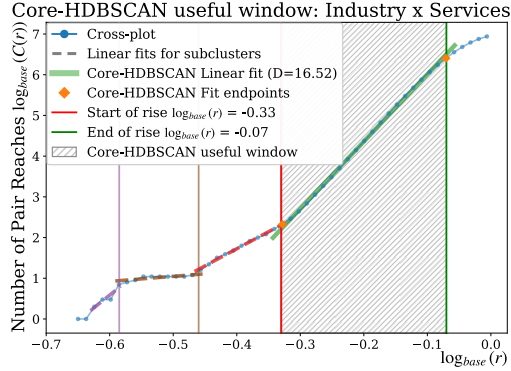


Fig. 3. Cross-plot for the Industry $\times$ Services pair, showing the Core-HDBSCAN useful window (hatched region) bounded by the rise onset and offset (red/green vertical lines), subcluster linear fits (dashed lines), and boundary markers (colored vertical lines).

Figure 3 details the cross-plot for the Industry $\times$ Services pair. This example exhibits a heterogeneous distribution (Property 2), where distinct slopes and plateaus, shown with vertical markers and piecewise dashed segments, reveal the formation of internal subclusters before the dominant macroscopic growth. The extraction of the optimal window follows Rule 1, which bypasses the early

micro-cluster structures and defines the start of the useful region (red vertical line) at the onset of the final steep rise. This ensures that the measured spatial correlation reflects the interaction between the macroscopic sets. Finally, the linear fit within the selected region provides the fractal correlation dimension (D_2), estimated as the slope of the regression line:

$$D_2 = \frac{dy}{dx} \approx \frac{\Delta \log(\text{Cross}_{A,B}(r))}{\Delta \log(r)} \quad (3)$$

As described, the macroscopic correlation window begins at the point of maximum topological acceleration (x_{ref}), where $y''(x)$ reaches its peak.

In this example, the extracted range $[-0.33, -0.07]$ spans the steepest growth region and yields an optimal spatial threshold for $\epsilon^* = 10^{-0.33} \approx 0.468$. Using this value as the `cluster_selection_epsilon` parameter provides HDB-SCAN with a deterministic geometric boundary, reducing fragmentation and aligning the clustering with the macroscopic structure of the semantic data.

4 Experimental Results

4.1 Dataset and Domain Partitioning

The PIESP dataset: the Survey of Announced Investments in the State of São Paulo (*Pesquisa de Investimentos Anunciados no Estado de São Paulo*) was obtained from the SEADE website¹. It comprises approximately 14.5 thousand investment records in the State of São Paulo over 13 years (2012–2025). It tracks productive investments, including companies, localities, descriptions, economic sectors, and announcement dates. For our experiments, we used two attributes: the textual investment description and the sector description (excluding the minor Agribusiness category), which serves as the ground-truth label for clustering evaluation.

Dataset Distribution: The dataset encompasses investment records across five primary economic sectors: Services (6,792 records; logistics, delivery, multi-use facilities), Commerce (4,134 records; retail, wholesale, distribution), Infrastructure (1,990 records; water, energy, sanitation), Industry (1,540 records; manufacturing, factories, tech development), and Agribusiness (134 records; agriculture, rural infrastructure).

The dataset exhibits a natural class imbalance across sectors. We preserved the original distribution without synthetic balancing (e.g., over/undersampling) to evaluate the robustness of the topological ϵ^* extraction under realistic conditions, leaving imbalance adjustments for future work. To embed the Portuguese corpus, raw texts were processed via the sentence-transformers library using the Bertimbau model(`neuralmind/ bert-base-portuguese-cased`)², which generated

¹ <https://repositorio.seade.gov.br/dataset/seade-investimentos/resource/3ee8eb9b-a3b5-4d53-925f-71e5dabb263c>

² <https://huggingface.co/neuralmind/bert-base-portuguese-cased>

768-dimensional dense embeddings, providing the continuous spatial representation needed to compute Euclidean distances for our fractal-based cross-plot methodology.

4.2 Experimental Evaluation and Clustering Results

With the dataset embedded in a high-dimensional space and the optimal threshold (ϵ^*) extracted via cross-cloud, we evaluate our methodology by comparing baseline HDBSCAN (H_{def} , using default parameters with no active spatial constraint) against our proposed model (*Core*, incorporating ϵ^* as a deterministic cluster extraction boundary).

$$H_{\text{def}} = \text{HDBSCAN}(\text{min_cluster_size} = 5, \text{cluster_selection_epsilon} = 0.0) \quad (4)$$

$$\text{Core} = \text{HDBSCAN}(\text{min_cluster_size} = 5, \text{cluster_selection_epsilon} = \epsilon^*) \quad (5)$$

The specific ϵ^* values for each pair, derived from their fractal acceleration points (x_{ref}), are detailed in Table 1. We evaluate how this structural restriction affects the ratio of valid clustered points versus noise, factoring in the dataset’s distribution. Table 2 summarizes the comparative performance across the analyzed economic domains.

Table 1. Spatial threshold (ϵ^*) calibration parameters, derived by converting the logarithmic rise onset (x_{ref}) back to the original distance scale.

Pair	x_{ref}	ϵ^*
Commerce $\times$ Infrastructure	-0.45	0.355
Commerce $\times$ Industry	-0.39	0.407
Commerce $\times$ Services	-0.50	0.316
Infrastructure $\times$ Industry	-0.35	0.447
Infrastructure $\times$ Services	-0.34	0.457
Industry $\times$ Services	-0.33	0.468

Table 2. Comparative clustering performance of HDBSCAN (H_{def}) $\times$ *Core* across sector pairs, detailing instance counts, formed clusters, and noise percentages.

Pair	Total Points	H_{def}	H_{def}	<i>Core</i>	<i>Core</i>
		Clusters	Noise (%)	Clusters	Noise (%)
Commerce $\times$ Infrastructure	6124	98	72.24	76	65.45
Commerce $\times$ Industry	5674	80	76.84	40	59.43
Commerce $\times$ Services	10926	168	79.75	144	77.24
Infrastructure $\times$ Industry	3530	32	81.93	28	81.16
Infrastructure $\times$ Services	8782	118	82.60	55	65.67
Industry $\times$ Services	8332	103	86.02	39	67.02

Table 2 shows that Core-HDBSCAN consistently reduces both noise and cluster counts relative to the baseline. By controlling spatial strictness, ϵ^* mitigates premature noise assignment and over-fragmentation, promoting denser, more cohesive clusters. Their semantic quality was assessed using the purity metric.

Table 3. Clustering Purity Comparison

Pair	H_{def}	Purity %	Core Purity%
Commerce $\times$ Infrastructure	99.59		99.62
Commerce $\times$ Industry	98.71		98.78
Commerce $\times$ Services	97.83		96.90
Infrastructure $\times$ Industry	98.28		98.35
Infrastructure $\times$ Services	98.69		98.77
Industry $\times$ Services	99.40		97.34

As detailed in Table 3, the proposed method improved semantic purity in four of the six analyzed pairs. The exceptions were ‘Commerce $\times$ Services’ (decreasing from 97.83% to 96.90%) and ‘Industry $\times$ Services’ (from 99.40% to 97.34%). Consequently, the overall average purity dropped slightly by 0.46% (from 98.75% to 98.29%). This trade-off is highly acceptable, as the topological ϵ^* maintains strong semantic cohesion while expanding cluster boundaries to capture thousands of instances previously discarded as noise.

To complement absolute purity and assess agreement between predicted partitions and ground-truth labels, we adopted pairwise counting metrics: precision, recall, and F1-score. These metrics evaluate clustering quality across all unordered instance pairs, treating noise points as misclassifications by verifying whether each pair is placed in the same cluster in both partitions.

Pairwise precision and recall measure the proportion of correctly co-clustered pairs relative to all predicted and true within-class pairs, respectively, balanced by the pairwise F1-score. To prevent macro-averaging from overemphasizing minority classes, we used weighted averaging to scale metric contributions by class frequencies. This accounts for sector-pair imbalances and ensures a representative performance evaluation (Table 4).

Table 4. Pairwise counting evaluation.

Pair	Precision	Recall	F1-Score
Commerce $\times$ Infrastructure	0.996 0.996	0.276 0.344	0.432 0.508
Commerce $\times$ Industry	0.982 0.982	0.229 0.401	0.359 0.533
Commerce $\times$ Services	0.981 0.969	0.198 0.221	0.325 0.354
Infrastructure $\times$ Industry	0.974 0.975	0.178 0.185	0.294 0.303
Infrastructure $\times$ Services	0.989 0.987	0.172 0.339	0.291 0.504
Industry $\times$ Services	0.992 0.972	0.139 0.321	0.243 0.468

Finally, to visually characterize topological boundaries and identify semantic overlap, we generated the confusion matrix for the calibrated extractions,

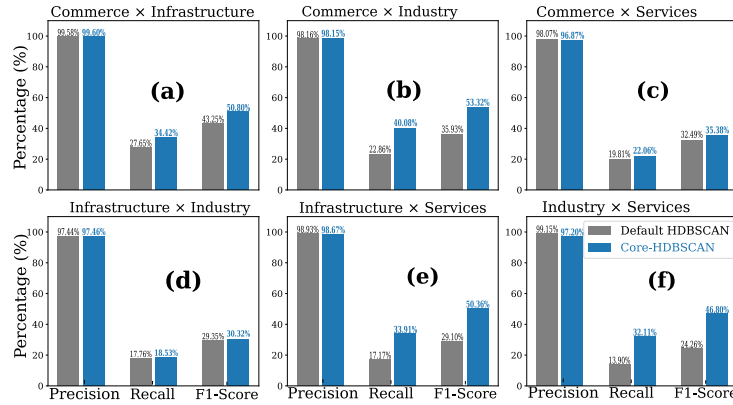


Fig. 4. Comparison of clustering metrics (Precision, Recall, F1-Score) between default HDBSCAN (gray) and Core-HDBSCAN (blue) across sector pairs. Subplots display percentages for each pair, highlighting differences in classification quality.

mapping true classes (rows) against predicted classes (columns). Unclassified points are explicitly penalized as "noise." Because the ground truth covers the entire semantic space with no outliers, assigning a point as noise represents an error in mapping the topological manifold. This penalty ensures the evaluation reflects each technique’s ability to capture structural completeness rather than just internal cluster homogeneity.

Complementing the global metrics in Figure 4, the confusion matrices highlight that our method matches or outperforms the baseline along the main diagonal (true positives) in 5 of the 6 evaluated pairs. The exception occurs in the Industry vs. Services pair (f), where the calibrated threshold significantly increases correct classifications for Services but slightly reduces them for Industry. This behavior naturally reflects the dataset’s asymmetric densities; under class imbalance, the larger manifold inherently assimilates boundary points. Rather than forcing a compressed fit like traditional dimensionality reduction, the fractal dimension threshold adapts to each domain’s specific topology, driving the overall improvements observed in the weighted F1-Score.

5 Conclusions

This paper introduces a new approach for identifying an optimal spatial threshold from pairwise dataset correlations. By deriving the cut-off parameter ϵ^* through fractal correlation analysis, our method reveals the intrinsic dimensional structure of the semantic space and directly calibrates density-based algorithms, bypassing empirical tuning. Empirical results validate the structural consistency of this extracted threshold, yielding an average relative F1-Score improvement of **37.35%** over the baseline configuration.

Core-HDBSCAN is particularly effective for hierarchical algorithms, like HDBSCAN, where ϵ^* acts as a spatial constraint for dendrogram pruning that mit-

igates over-fragmentation while preserving domain structure. Future work will investigate the scalability of this topological mapping across different embedding architectures and textual corpora.

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